1) Event Communication Security

a) Reconstructed Encrypted Message.

Using a classical caesar cipher with a of fixed shift of 4 positions forward.

$C = P + 4 \pmod{26}$.

Plain text	Ciphertext
M	Q
E	I
E	I
T	X
A	E
T	X
N	R
O	S
O	S
N	R

MEET AT NOON → QIIX EX RSSR

b) Decryption Rule for Authorised Members.

→ To retrieve the original message, an authorized member reverses the process by applying a shift of 4 positions backward. ($P = C - 4 \pmod{26}$) to each letter of the ciphertext

2) Intercepted Suspicious Message.

a) Most likely original Message.

Plaintext : THIS IS A SECRET

b) Justification & Analysis.

-> Cipher characteristics : This message is encrypted using the monoalphabetic Atbash cipher, which mirror the alphabet completely (A↔Z, B↔Y, C↔X, etc..,).

-> Pattern Breakdown

- The single-letter word Z map directly to A which is the most common single letter word in english grammar.
- Repeat structure RH maps to IS, matching perfectly with the structure of standard sentence.
- Substituting the letter based on the mirror rule (Position - 27 - Position) fully decrypts the sentence to :

GSRH RH Z HVXIVG -> THIS IS A SECRET

3) Scenario : Secure Message System

a) Step - by-Step Encryption Process.

This scenario uses a polyalphabetic vegnera cipher The key word "KEY" is Repeated to match the length of the plain text HELLO.

(PT)			Cipher text
H	-> H + K	-> 7 + 10 = 17	-> R
E	-> E + E	-> 4 + 4 = 8	-> I
L	-> L + Y	-> 11 + 24 = 35 mod 26 => 9.	-> J
L	-> L + K	-> 11 + 10 = 21	-> V
O	-> O + E	-> 14 + 4 = 18	-> S

HELLO -> RIJVS

b) Influnce of Repeating Keyword.

The repeated keyword eliminates simple letter frequency patter As demonstrated above the identical Consecutive plaintext L and L are transformed in to completely different Cipher text letter (J and V) because they ar shifted by different letters of the keyword (Y and K).

4) Suspicious Digital Asset Investigation

a) Detection of Hidden data.

i) Statistical Stagandysis: Investigator can use a Chi - Square attack to analyse color frequency distribution. Natural image have smooth variations, where staganography introduces mathematical anomalies.

ii) File property & Hex Analysis: Investigator check for unusual file size compared to standard resolution or use Hex editors to search for structural irregularities like hidden data appened past the standard. EOF marker.

b) Technique to Strengthen Security.

i) Pre- encryption: Encryption the confidential data using a strong morden standard before embedded it. even if detected and extracted, the adversary only obtains unreadable random noise.

5) legacy Encryption in corporate Communication

a) The cipher text : WECRLTEERDSOEEFEAOC

AIVDEN has 25 characters and was encrypted using 3 level Rail fence.

→ Rail fence zigzag grid layout

Rail 1 → 7

Rail 2 → 12

Rail 3 → 6.

Rail 1 ⇒ WECRLTE

Rail 2 ⇒ ERDSOEEFEAOC

Rail 3 ⇒ AIVDEN

WEAR ED IS COVER

b) Transposition cipher only rearranges letters without changing them the letter frequency distribution remains identical to plain English text allowing quick identification